

Table 1: Explanatory Power of Theoretical Frameworks for Structural Phenomena in LLMs

Phenomenon	Matrix Structural Theory	Information Geometry	Statistical Mechanics	Topology	SQFT (Semantic QFT)
Rank Collapse	✓ Low-rank constraints	✓ Fisher rank	✓ Free energy minimization	✓ Dimensional degeneration	✓ Spectral reorganization
Covariance Cutting	✓ Off-diagonal suppression	✓ Fisher matrix block structure	✓ Correlation length decay	✓ Strata disconnection	✓ Gauge curvature
Modular Specialization	✓ Block decomposition	✓ Mixture singularities	✓ Order parameters	✓ Strata separation	✓ Semantic condensation
Scaling Law Curvature	✓ Spectral saturation	✓ Entropy saturation	✓ Free energy saturation	✓ Dimensional limits	? Not yet specified
Representation Redundancy	✓ Effective rank saturation	~ Information dimension	~ Entropy	×	×
Theoretical Fragmentation	✓ Structural unification	×	×	×	×